

JuliaDB: An Analytical Database that Performs Better than its Counterparts



Julia is a programming language that produces code which runs considerably faster than its counterparts. This article introduces JuliaDB – an analytical database that is entirely created in Julia. The various advantages of JuliaDB include JIT compilation, powerful parallelism and a super-fast CSV parser. So let's explore JuliaDB further.

Julia is a powerful programming language. The programming languages can be broadly grouped into two categories — those that are easy to build programs with but the code created runs comparatively slower (e.g., Python), and those with which building programs requires more time but the resultant code runs considerably faster (e.g., C++). Julia is aimed at combining the benefits of both these programming worlds. If you are new to Julia, there have been earlier articles (five, to be precise) on the subject in *Open Source For You* itself, which can be referred to at <https://www.opensourceforu.com/?s=Julia>.

JuliaDB is an analytical database. When compared with similar options such as Pandas, the features provided by JuliaDB are better. At the same time, JuliaDB is also comparatively faster.

The major features of JuliaDB are listed below:

- Just-in-time compilation
- Superior parallelism
- It can store many types of data.
- The UDF (user defined functions) are faster, as they are also JIT compiled.

- It has a super-fast CSV parser that can handle large files without a lag.

Comparing the performance of JuliaDB with other analytical databases

As per the official website of JuliaDB, the performance with respect to large CSV files is better. Figure 2 shows a comparison between JuliaDB and Pandas (source: <https://juliadb.org/>).

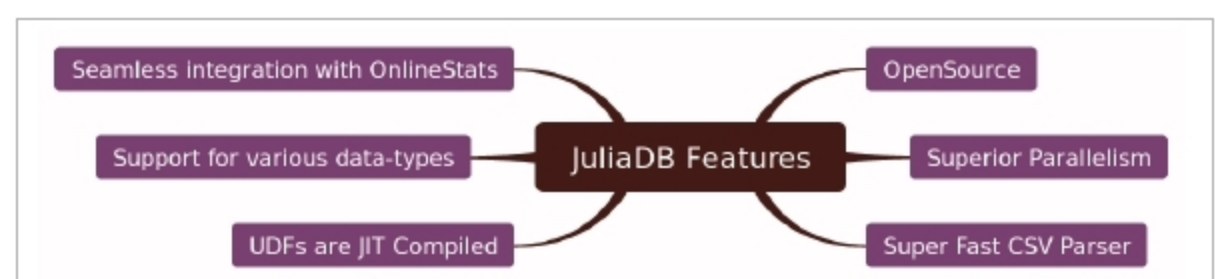


Figure 1: JuliaDB – Major features